

Richy Koshy

python engineer · genai, backend & automation

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“Conscientious ownership, earnest curiosity, and diligent follow-through.”

Corban Wells, VP of Technology Product Management, PRMI · full recommendation letter available on request

5+ years of Python development, plus **a year of industrial automation**. Shipped production GenAI, backend, and automation systems for a US mortgage originator: an enterprise Azure RAG platform, a custom Encompass REST API SDK, and 13 production automations across compliance and investor operations.

01 · work experience

Software Developer · Primary Residential Mortgage, Inc.

US mortgage originator (PRMI) · sole engineer owning requirements, architecture, build, UAT, and production

- **MAX AI, enterprise RAG platform.** Owned the full Python backend across three Azure services; now PRMI's primary knowledge-retrieval tool **organization-wide**.
 - **Retrieval:** hybrid vector + semantic ranking on Azure AI Search with HyDE query expansion; Azure OpenAI GPT-4o with ada-002 embeddings; one index per project, with multi-tenant access mapping in SQL Server behind Azure AD auth.
 - **Ingestion:** Durable Functions fan-out pipeline, idempotent via audit-table checkpointing with per-file failure isolation; local PDF partitioning and table extraction with unstructured (poppler + Tesseract), shipped as a Docker image on Azure Container Registry; Azure Document Intelligence for scans.
 - **Chat:** FastAPI service streaming NDJSON with metadata-first framing, so citations render before the first token.
- **Final Docs refactor.** Rebuilt a monolithic automation into config-driven dynamic dispatch (YAML → handlers), treating each investor as data, not code: **4 new investors onboarded in 1 month** vs. a full quarter before; onboarding cut from **2-4 weeks to days**; adopted as the template for 2 other initiatives.
- **Reconciliation engine.** Strategy pattern with one YAML config per investor; pandas merge on composite keys reconciles internal records against investor files; unmatched records get a fail-safe Encompass lookup, and true exceptions land in an operations review queue, so nothing is silently dropped.

“PRMI now treats investor expansion as a business decision, not a dev-capacity constraint.”

Corban Wells, VP Technology Product Management, PRMI
- **Encompass REST API SDK.** Internal Python package consumed by all 13 automations: OAuth2 token caching, concurrent paginated pipeline reads (ThreadPoolExecutor), transactional loan writes with a lock → write → read-back → unlock pattern that prevents lost updates on shared loans, eFolder document CRUD, disclosure tracking, and 429 retry with backoff.
- **13 production RPA automations** (Power Automate + Python) across ECOA adverse action, TRID, HMDA, MERS sync, final docs, HFA exceptions, Blend credit management, warehouse collateral, and suspense resolution across 9 investors; plus 3 OCR pipelines (Tesseract + Azure Form Recognizer) extracting credit data from PDF reports.
- **Delivery ownership.** Led BRD/PRD gathering, flow diagrams, first-call sign-offs, UAT coordination, and documentation for every delivered automation.

Software Developer · Antz AI Solutions

client: TechSpan Engineering (IoT)

- **HydroQ API.** FastAPI + SQLAlchemy REST API for remote water-quality monitoring with a generic CRUD layer over PostgreSQL and MSSQL; replaced **2-4 hours of manual onboarding per site** with API-driven workflows; full endpoint test coverage.

Industrial Automation Engineer · Goodyear South Asia Tyres

tyre manufacturing · PLC, drives, and controls

- Programmed and maintained PLCs and variable-frequency drives across tyre-curing and material-handling lines.
- Led breakdown troubleshooting on production-critical electrical systems, minimizing line downtime.
- Commissioned CTMS & BMS systems alongside Rockwell and Honeywell engineers.

02 · skills

GENAI / RAG

Azure OpenAI (GPT-4o, ada-002), Azure AI Search, hybrid vector + semantic retrieval, HyDE, ingestion pipelines

BACKEND

Python, FastAPI, asyncio, SQLAlchemy, Pydantic, REST API design, OAuth2, pytest

CLOUD & INFRA

Azure (Durable Functions, App Services, Container Registry, Blob/Table Storage), Docker, SQL Server, PostgreSQL

AUTOMATION

Power Automate, RPA, Encompass REST API, Blend API, Tesseract / Azure Form Recognizer OCR, Pandas

03 · github · richie-rk

rag-app-azure

enterprise Azure RAG app: hybrid search (vector + semantic + BM25), NDJSON streaming, MSAL SSO, FastAPI + Azure Functions + React

Bible-VerseFinder-AI

AI-powered Bible verse search engine with LLM summarization and full chapter reading

PyTeslaCoil

pure Python implementation of Java TeslaCoil

EncompassRest-RAG-Assistant

natural-language RAG over ICE Mortgage Encompass API docs, hybrid retrieval, multi-LLM

azure-search-chunk-viewer

Python web app for browsing and analyzing Azure AI Search index chunks

04 · education & awards

BE, Electrical Engineering

University of Mumbai

Best Paper

Intl. Conference on Advanced Trends in Engineering

1st Place

Tantragyan National Project Competition